INCH-POUND

MIL-C-24643/6D
14 March 1994
SUPERSEDING
MIL-C-24643/6C
29 May 1987

MILITARY SPECIFICATION SHEET

CABLE, ELECTRICAL, 600 VOLTS, AC., TYPE LSMDY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-24643.

REQUIREMENTS:

Qualification required.

Construction (watertight)

First - Copper conductor, uncoated, (see table I for size). Coated copper may be used at manufacturer's option where required to provide free stripping insulation.
Second - Separator may be used at manufacturer's option where required to provide free-stripping insulation.
* Third - Ethylene propylene rubber or cross-linked polyethylene insulation or at the manufacturer's option a composite with a primary insulation consisting of ethylene propylene rubber or cross-linked polyethylene. The composite layer may be substituted for up to 10 percent of the wall thickness (see table I for wall thickness). Standard identification code shall be applied by method 1 or method 3. If identification method 1 is used, insulation color shall be black.
Fourth - Nineteen conductors, cabled together with a lay not greater than 24 times the pitch diameter of the layer. Cabling sequence to be consecutive, starting with no. 1 from center outward. Fillers shall be used as
* necessary to make a firm, well-rounded assembly.
Fifth - Binder tape applied helically with overlap.
Sixth - Cross-linked polyolefin jacket.
Seventh- Braided metal armor.
Eighth - Binder tape applied helically with overlap.
* Ninth - Cross-linked polyolefin jacket. (see table I for wall thickness).

AMSC N/A FSC 6145

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Table I. Details

| Military part no. M24643/6 | Type and size | Conductors | Insulation thickness minimum average | Inner cable Jacket Thickness min avg | Diameter over armor | Cable Jacket Thickness minimum average | Overall Diameter | | Conductor resistance per 1000 feet | Insulation resistance per 1000 feet | Cold bending mandrel diameter |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Size AWG | (inch) | (inch) | nominal (inch) | (inch) | minimum (inch) | maximum (inch) | maximum (ohms) | minimum (megohms) | maximum (inch) |
| -01AN | LSMDY-6 | 12(class B) | 0.028 | 0.050 | 1.000 | 0.060 | 1.120 | 1.190 | 1.73 | 100 | 15 |
| -02AN | LSMDY-14 | 9(class B) | .040 | .050 | 1.380 | .060 | 1.500 | 1.570 | .868 | 100 | 21 |
| -03AN | LSMDY-23 | 7(class B) | .052 | .085 | 1.710 | .075 | 1.880 | 1.960 | .598 | 100 | 25 |
| -04AN | LSMDY-40 | 4(class C) | .052 | .065 | 1.960 | .075 | 2.150 | 2.240 | .273 | 90 | 29 |
| -05AN | LSMDY-60 | 2(class D) | .052 | .065 | 2.250 | .075 | 2.425 | 2.525 | .172 | 75 | 33 |



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EXAMINATION AND TESTS:

| Basic electrical: | Requirements |
|---|---|
| Conductor Resistance - ohms/1000 feet at 25°C, maximum | (see table I) |
| Voltage withstand - volts, root mean square, minimum | |
| Conductor to conductor | 2500 |
| Insulation resistance - megohms/1000 feet, minimum | (see table I) |
| Conductor continuity | No failure |
| **Group A:** | |
| Visual and dimensional | No failure |
| Watertightness - see MIL-C-24643 for limits of water leakage | No failure |
| **Group B:** | |
| Cold bending, cable - (see table I for mandrel diameter) | No damage |
| * Cross-linked proof test (percent, maximum) | |
| Insulation | 50 |
| Jacket (When tested at 200°C) | 50 |
| Drip - 95 ± 1°C | Zero |
| Tear - pounds per inch thickness, minimum (ASTM D 470) | 35 |
| Physicals (unaged) | |
| Insulation | |
| Ethylene propylene rubber | |
| Tensile strength - $lb/in^2$, minimum | 700 |
| Elongation - percent, minimum | 250 |
| Cross-linked polyethylene | |
| Tensile strength - $lb/in^2$, minimum | 1800 |
| Elongation - percent, minimum | 250 |
| Jacket (cable) | |
| Tensile strength - $lb/in^2$, minimum. | 1300 |
| Elongation - percent, minimum | 160 |
| **Group C:** | |
| Physicals (aged) | |
| Insulation | |
| Ethylene propylene | |
| Air oven | |
| Tensile strength - percent of unaged, minimum | 75 |
| Elongation - percent of unaged minimum | 75 |
| Cross-linked polyethylene | |
| Air oven | |
| Tensile strength - percent of unaged, minimum | 80 |
| Elongation - percent of unaged minimum | 80 |



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EXAMINATION AND TESTS (continued):

| | | Requirements |
|---|---|---|
| | Jacket (cable) | |
| | Air oven | |
| | Tensile strength - percent of unaged, minimum | 60 |
| | Elongation - percent of unaged minimum | 60 |
| * | Hot oil immersion | |
| | Tensile strength - percent of unaged, minimum | 50 |
| | Elongation - percent of unaged minimum | 50 |
| | Shrinkage | No failure |
| * | Permanence of printing (jacket) - cycles, minimum | 125 |
| * | Permanence of printing (conductor) - cycles, minimum | 25 |
| | Heat distortion - percent of unaged, maximum | 30 |
| * | Cable sealant removability | No failure |
| * | Group D: | |
| | Flame propagation (cable) | No failure |

QUALIFICATION INSPECTION:

* Qualification inspection shall include basic electrical, all of groups A, B, C and D, plus the following:

| | Requirements |
|---|---|
| Aging and compatibility (cable) (125 ±5°C.) | No failure |
| Abrasion resistance (jacket) - scrapes, minimum | 75 |
| Acid gas equivalent - percent, maximum | |
| Jacket | 2 |
| Fillers | 2 |
| Insulation | 18 |
| Halogen content - percent, maximum | |
| Jacket | 0.2 |
| Fillers | 0.2 |
| Immersion (jacket) | |
| Tensile strength - percent of unaged, minimum | 50 |
| Elongation - percent of unaged, minimum | 50 |
| Smoke index, maximum | |
| Jacket | 25 |
| * Fillers | 45 |
| Insulation | 45 |
| Toxicity index, maximum | |
| Jacket | 5 |
| Fillers | 5 |
| Insulation | 1.5 |
| Durometer (jacket) - (type A) hardness, minimum | 80 |
| Weathering (jacket) | No failure |



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UNIT ORDERING LENGTH: 1000 feet nominal length for sizes 6, 14 and 23.
500 feet nominal for sizes 40 and 60.

NOTE: The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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Review Activities:
Army - AV, CR, ER, ME, AR, AL
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Preparing Activity:
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Agent:
DLA-IS

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